

Studies on Domain Walls, Cosmic Strings, and Their Gravitational Wave Signatures

Zhao-Huan Yu (余钊焕)

School of Physics, Sun Yat-Sen University

<https://yzhxxzxy.github.io>

Based on Qing-Quan Zeng, Xi He, ZHY, Jiaming Zheng, arXiv:2501.10059, PRD
Shi-Qi Ling, ZHY, arXiv:2502.16576, CPC



The 6th Workshop on Frontiers of Particle Physics

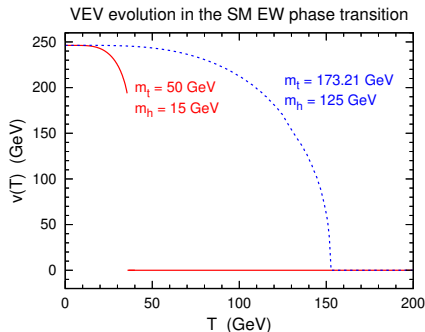
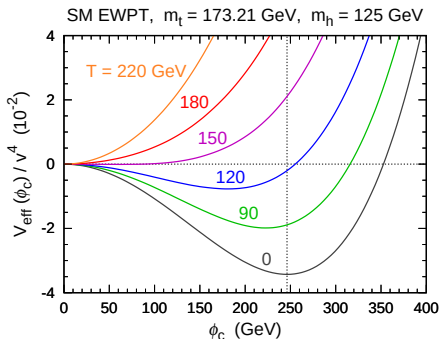
Jilin University, July 17, 2025



Cosmological Phase Transition

🔥 **Spontaneously broken symmetries** in field theories can be **restored** at **sufficiently high temperatures** due to **thermal corrections** to the **effective potential**

☁️ In the history of the Universe, **spontaneous symmetry breaking** manifests itself as a **cosmological phase transition**



Topological Defects



Consider that **some scalar fields** acquire nonzero **vacuum expectation values** (VEVs), which **break** a **symmetry group** G to a **subgroup** H



The **manifold** consisting of all **degenerate vacua** is the **coset space** G/H



The **topology** of the **vacuum manifold** G/H can be characterized by its **n -th homotopy group** $\pi_n(G/H)$, which is constituted by the homotopy classes of the mappings from an **n -dimensional sphere** S^n into G/H



A **nontrivial** $\pi_n(G/H)$ leads to **topological defects** [Kibble, J. Phys. A9 (1976) 1387], as commonly predicted in **grand unified theories**

Topological Defects



Consider that **some scalar fields** acquire nonzero **vacuum expectation values** (VEVs), which **break** a **symmetry group** G to a **subgroup** H



The **manifold** consisting of all **degenerate vacua** is the **coset space** G/H



The **topology** of the **vacuum manifold** G/H can be characterized by its **n -th homotopy group** $\pi_n(G/H)$, which is constituted by the homotopy classes of the mappings from an **n -dimensional sphere** S^n into G/H



A **nontrivial** $\pi_n(G/H)$ leads to **topological defects** [Kibble, J. Phys. A9 (1976) 1387], as commonly predicted in **grand unified theories**



Nontrivial $\pi_0(G/H)$: two or more disconnected components



Domain walls (2-dim topological defects)



Nontrivial $\pi_1(G/H)$: incontractable closed paths



Cosmic strings (1-dim topological defects)



Nontrivial $\pi_2(G/H)$: incontractable spheres



Monopoles (0-dim topological defects)



$$\pi_0(G/H) = \mathbb{Z}_2$$



$$\pi_1(G/H) = \mathbb{Z}$$

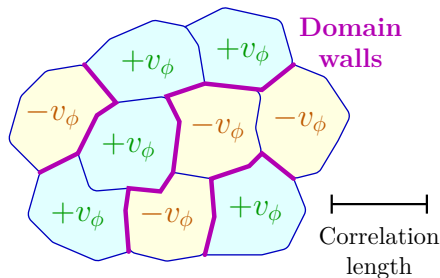
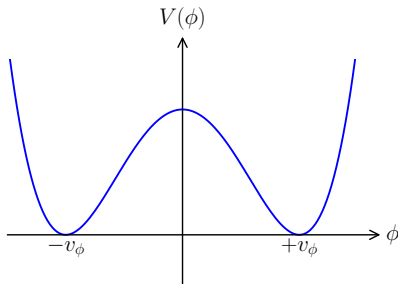
Domain Walls

🌀 **Domain walls (DWs)** are **two-dimensional topological defects** which could be formed when a **discrete symmetry** of the **scalar potential** is **spontaneously broken** in the early Universe

▮ They are **boundaries** separating spatial regions with different **degenerate vacua**

🚫 **Stable DWs** are thought to be a **cosmological problem** [Zeldovich, Kobzarev, Okun, Zh.Eksp.Teor.Fiz. **67** (1974) 3]

⚠️ As the Universe expands, the **DW energy density** decreases **slower** than radiation and matter, and would soon **dominate** the total energy density



Collapsing Domain Walls



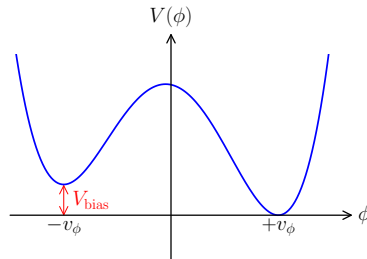
It is **allowed** if **DWs collapse** at a very early epoch [Vilenkin, PRD **23** (1981) 852; Gelmini, Gleiser, Kolb, PRD **39** (1989) 1558; Larsson, Sarkar, White, hep-ph/9608319, PRD]



Such **unstable DWs** can be realized if the **discrete symmetry** is **explicitly broken** by a **small potential term** that gives an **energy bias** V_{bias} among the minima of the potential



The bias induces a **volume pressure force** acting on the DWs that leads to their collapse



Collapsing DWs can produce significant **GWs** [Preskill *et al.*, NPB 363 (1991) 207; Gleiser, Roberts, astro-ph/9807260, PRL; Hiramatsu, Kawasaki, Saikawa, 1002.1555, JCAP]



A **stochastic gravitational wave background (SGWB)** would be formed and remain to the present time

Spontaneously Broken Z_2 Symmetry

 We study the dynamics of **DWs** formed through the **spontaneous breaking** of an **approximate Z_2 symmetry** in a **scalar field ϕ** , focusing on the influence of **quantum** and **thermal corrections** induced by a **Z_2 -violating Yukawa coupling** to **Dirac fermions f** in the **thermal bath** [QQ Zeng, X He, **ZHY**, JM Zheng, 2501.10059, PRD]

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + i \bar{f} \gamma^\mu \partial_\mu f - m_f \bar{f} f - y \phi \bar{f} f - V_0(\phi)$$

$$V_0(\phi) = -\frac{1}{2} \mu_\phi^2 \phi^2 + \frac{1}{3} \mu_3 \phi^3 + \frac{1}{4} \lambda_\phi \phi^4, \quad \mu_\phi^2 > 0, \quad \lambda_\phi > 0$$

 The small **couplings y** and **μ_3 explicitly violate** the **Z_2 symmetry $\phi \rightarrow -\phi$**

 Considering the **Coleman-Weinberg correction $V_{\text{CW}}(\phi)$** and **finite-temperature correction $V_{\text{T}}(\phi, T)$** at one-loop level, the **effective potential** becomes

$$V(\phi, T) = V_0(\phi) + V_{\text{CW}}(\phi) + V_{\text{T}}(\phi, T)$$

 The **vacuum expectation value (VEV) v_ϕ** of ϕ corresponds to

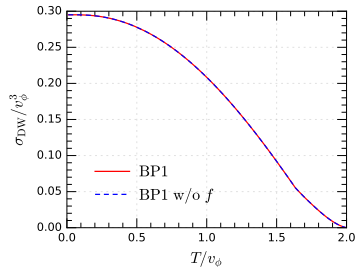
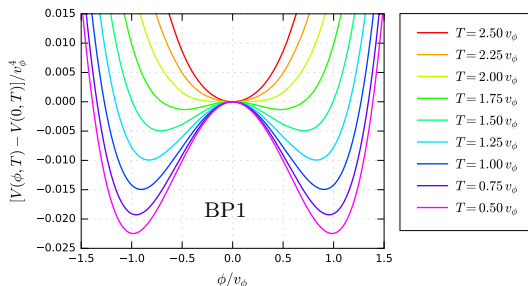
$$\left. \frac{\partial}{\partial \phi} [V_0(\phi) + V_{\text{CW}}(\phi)] \right|_{\phi=v_\phi} = 0$$

Potential Evolution and DW Tension

☂ By solving the equation of motion for the **DW solution** and integrating the energy density, the **DW tension** σ_{DW} , i.e., **energy per unit area**, can be obtained

☁ We choose **three benchmark points** (BPs) to highlight remarkable features

	v_ϕ [GeV]	μ_3/v_ϕ	λ_ϕ	y	m_f/v_ϕ
BP1	3×10^9	-10^{-17}	0.1	4.65×10^{-5}	4×10^{-5}
BP2	6×10^4	-10^{-27}	0.1	2.5×10^{-8}	5×10^{-7}
BP3	1.5×10^{11}	-3.645×10^{-13}	0.1	3×10^{-4}	4×10^{-4}



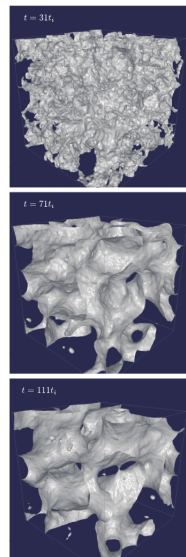
Evolution of Domain Walls

After DWs are created, the **tension** σ_{DW} acts to **stretch** them up to the **horizon size** if the **friction** F_f is **small**, and they would enter the **scaling regime** with **energy density** $\rho_{\text{DW}} = \frac{\mathcal{A}\sigma_{\text{DW}}}{t}$

$\mathcal{A} \approx 0.8 \pm 0.1$ is a numerical factor given by lattice simulation

This implies that DWs are **diluted more slowly** than **radiation** ($\rho_r \propto t^{-2}$) and **matter** ($\rho_m \propto t^{-3/2}$) as the Universe expands

If DWs are **stable**, they would soon **dominate** the evolution of the Universe, **conflicting** with cosmological observations



[Hiramatsu *et al.*, 1002.1555]

Evolution of Domain Walls

After DWs are created, the **tension** σ_{DW} acts to **stretch** them up to the **horizon size** if the **friction** F_f is **small**, and they would enter the **scaling regime** with **energy density** $\rho_{\text{DW}} = \frac{\mathcal{A}\sigma_{\text{DW}}}{t}$

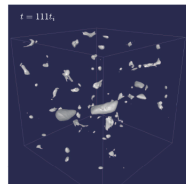
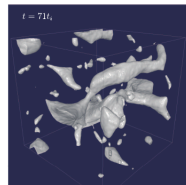
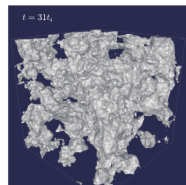
$\mathcal{A} \approx 0.8 \pm 0.1$ is a numerical factor given by lattice simulation

This implies that DWs are **diluted more slowly** than **radiation** ($\rho_r \propto t^{-2}$) and **matter** ($\rho_m \propto t^{-3/2}$) as the Universe expands

If DWs are **stable**, they would soon **dominate** the evolution of the Universe, **conflicting** with cosmological observations

However, the **potential bias** $V_{\text{bias}}(T) = V(\phi_-, T) - V(\phi_+, T)$ between the false and true vacua ϕ_- and ϕ_+ provides a **pressure** $p_V(T) \sim V_{\text{bias}}(T)$ acting on the DWs, against the **tension force per unit area** $p_T(T) \sim \rho_{\text{DW}}(T)$

This makes the **DWs collapse** and the **false vacuum domains shrink**

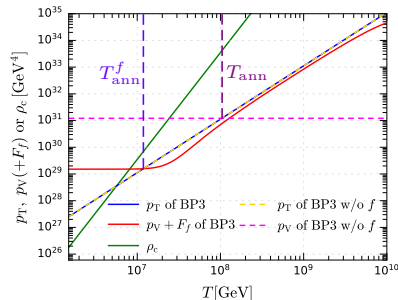
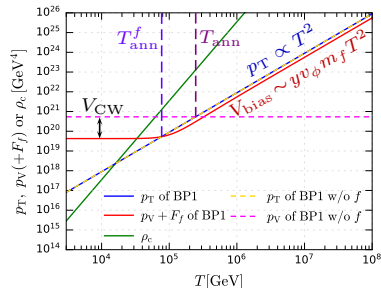
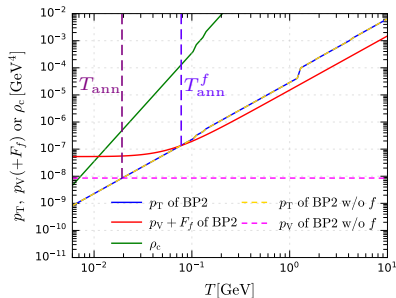
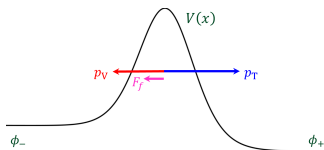


[Hiramatsu *et al.*, 1002.1555]

Annihilation Temperature

The **domain walls collapse** at the **annihilation temperature** T_{ann} when

$$p_V(T_{\text{ann}}) + F_f(T_{\text{ann}}) \simeq p_T(T_{\text{ann}})$$



SGWB Spectrum from Collapsing DWs



The **SGWB spectrum** is commonly characterized by $\Omega_{\text{GW}}(f) = \frac{f}{\rho_c} \frac{d\rho_{\text{GW}}}{df}$



ρ_{GW} is the **GW energy density**, and ρ_c is the critical energy density



The SGWB from **collapsing DWs** can be estimated by **numerical simulations**

[Hiramatsu, Kawasaki, Saikawa, 1002.1555, 1309.5001, JCAP]



The **present SGWB spectrum** induced by collapsing DWs can be evaluated by

$$\Omega_{\text{GW}}(f)h^2 = \Omega_{\text{GW}}^{\text{peak}}h^2 \times \begin{cases} \left(\frac{f}{f_{\text{peak}}}\right)^3, & f < f_{\text{peak}} \\ \frac{f_{\text{peak}}}{f}, & f > f_{\text{peak}} \end{cases}$$

$$\Omega_{\text{GW}}^{\text{peak}}h^2 = 7.2 \times 10^{-18} \tilde{\epsilon}_{\text{GW}} \mathcal{A}^2 \left[\frac{g_{*s}(T_{\text{ann}})}{10} \right]^{-4/3} \left[\frac{\sigma_{\text{DW}}(T_{\text{ann}})}{1 \text{ TeV}^3} \right]^2 \left(\frac{T_{\text{ann}}}{10 \text{ MeV}} \right)^{-4}$$

$$f_{\text{peak}} = 1.1 \times 10^{-9} \text{ Hz} \left[\frac{g_{*s}(T_{\text{ann}})}{10} \right]^{1/2} \left[\frac{g_{*s}(T_{\text{ann}})}{10} \right]^{-1/3} \frac{T_{\text{ann}}}{10 \text{ MeV}}$$

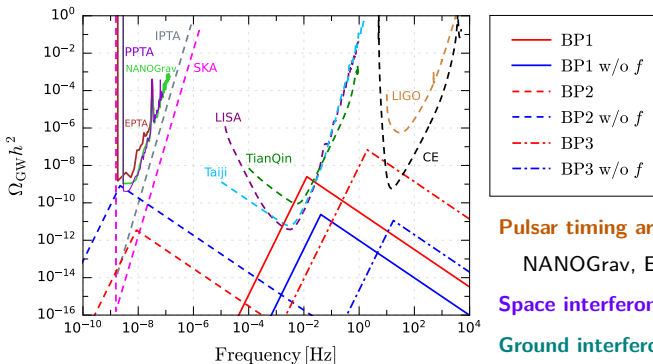


$\tilde{\epsilon}_{\text{GW}} = 0.7 \pm 0.4$ is derived from numerical simulation

SGWB Spectra with and without the Fermion

🎯 A **decrease** of T_{ann} by **one** order of magnitude would **increase** $\Omega_{\text{GW}}^{\text{peak}} h^2$ by **four** orders of magnitude and **decrease** f_{peak} by **one** order of magnitude

🔍 The **differences** between the **SGWB spectra** predicted by the scenarios **with** and **without the fermion** could potentially be verified by **future GW experiments**



Pulsar timing arrays (PTAs):

NANOGrav, EPTA, PPTA, IPTA, SKA

Space interferometers: LISA, TianQin, Taiji

Ground interferometers: LIGO, CE

Cosmic Strings from U(1) Gauge Symmetry Breaking

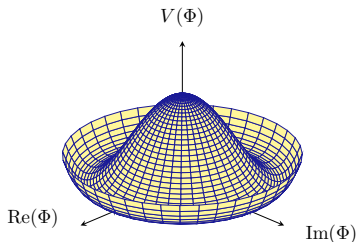
🧠 Consider the **Abelian Higgs model** with a **complex scalar field** Φ

$$\mathcal{L} = (D^\mu \Phi)^\dagger (D_\mu \Phi) - V(\Phi) - \frac{1}{4} X^{\mu\nu} X_{\mu\nu}, \quad V(\Phi) = -\mu_\phi^2 |\Phi|^2 + \frac{\lambda_\Phi}{2} |\Phi|^4$$

🟢 The covariant derivative of Φ is $D_\mu \Phi = (\partial_\mu - iq_\Phi g_X X_\mu) \Phi$

🧚 The field strength tensor of the **U(1)_X gauge field** X^μ is $X_{\mu\nu} = \partial_\mu X_\nu - \partial_\nu X_\mu$

🧸 Assume a **Mexican-hat potential** $V(\Phi)$ with **degenerate vacua** $\langle \Phi \rangle = v_\Phi e^{i\varphi} / \sqrt{2}$



Cosmic Strings from U(1) Gauge Symmetry Breaking

Consider the **Abelian Higgs model** with a **complex scalar field** Φ

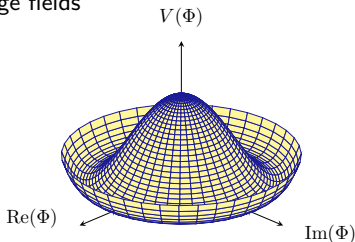
$$\mathcal{L} = (D^\mu \Phi)^\dagger (D_\mu \Phi) - V(\Phi) - \frac{1}{4} X^{\mu\nu} X_{\mu\nu}, \quad V(\Phi) = -\mu_\phi^2 |\Phi|^2 + \frac{\lambda_\Phi}{2} |\Phi|^4$$

The covariant derivative of Φ is $D_\mu \Phi = (\partial_\mu - iq_\Phi g_X X_\mu) \Phi$

The field strength tensor of the **U(1)_X gauge field** X^μ is $X_{\mu\nu} = \partial_\mu X_\nu - \partial_\nu X_\mu$

Assume a **Mexican-hat potential** $V(\Phi)$ with **degenerate vacua** $\langle \Phi \rangle = v_\Phi e^{i\varphi} / \sqrt{2}$

The **spontaneous breaking** of the **U(1)_X gauge symmetry** in the early Universe would induce **cosmic strings (CSs)**, which are concentrated with energies of scalar and gauge fields



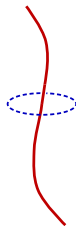
Degenerate vacua

$$v_\Phi e^{i\varphi} / \sqrt{2}$$

$$\varphi = \varphi + 2\pi n$$

$n \neq 0$ leads to

cosmic strings



Cosmic String Tension

📖 A **network** of **cosmic strings** would be formed in the early Universe after the spontaneous breaking of the $U(1)_X$ gauge symmetry

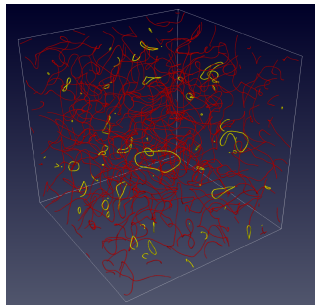
🧵 The **tension** of **cosmic string** μ (energy per unit length) can be estimated as

$$\mu \simeq \begin{cases} 1.19\pi v_\Phi^2 b^{-0.195}, & 0.01 < b < 100, \\ \frac{2.4\pi v_\Phi^2}{\ln b}, & b > 100, \end{cases} \quad b \equiv \frac{2q_\Phi^2 g_X^2}{\lambda_\Phi}$$

[Hill, Hodges, Turner, PRD **37**, 263 (1988)]

💥 As $\mu \propto v_\Phi^2$, a **high symmetry-breaking scale** v_Φ would lead to cosmic strings with **high tension**

💡 Denoting G as the **Newtonian constant of gravitation**, the **dimensionless quantity** $G\mu$ is commonly used to describe the **tension** of cosmic strings



[Kitajima, Nakayama, 2212.13573, JHEP]

Gravitational Waves from Cosmic Strings



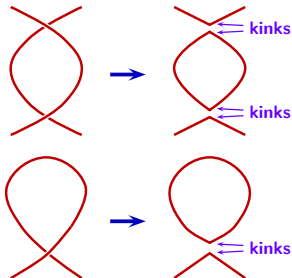
According to the analysis of string dynamics, the **intersections** of **long strings** could produce **closed loops**, whose size is smaller than the Hubble radius



Cosmic string loops could further fragment into **smaller loops** or reconnect to **long strings**



Loops typically have localized features called “**cusps**” and “**kinks**”



Gravitational Waves from Cosmic Strings

🏐 According to the analysis of string dynamics, the **intersections** of **long strings** could produce **closed loops**, whose size is smaller than the Hubble radius

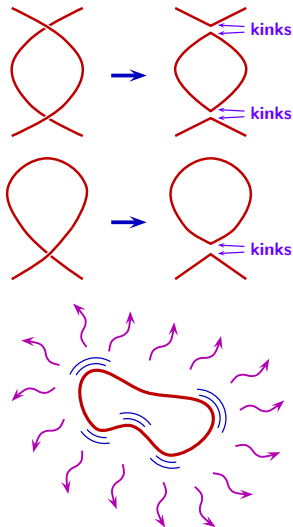
🏏 **Cosmic string loops** could further fragment into **smaller loops** or reconnect to **long strings**

🏏 Loops typically have localized features called **“cusps”** and **“kinks”**



📡 The **relativistic oscillations** of the **loops** due to their **tension** emit **Gravitational Waves (GWs)**, and the loops would **shrink** because of **energy loss**

🔔 Moreover, the **cusps** and **kinks** propagating along the loops could produce **GW bursts** [Damour & Vilenkin, gr-qc/0004075, PRL]



Power of Gravitational Radiation



At the **emission time** t_e , a **cosmic string loop** of **length** l emits GWs with **frequencies** $f_e = \frac{2n}{l}$



$n = 1, 2, 3, \dots$ denotes the **harmonic modes** of the loop oscillation



Denoting P_n as the **power** of **gravitational radiation** for the harmonic mode n in units of $G\mu^2$, the total power is given by $P = G\mu^2 \sum_n P_n$



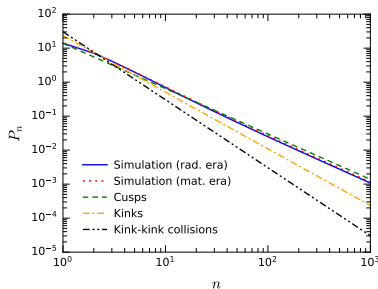
According to the **simulation** of **smoothed cosmic string loops** [Blanco-Pillado & Olum, 1709.02693, PRD], P_n for loops in the **radiation** and **matter** eras are obtained



The **total dimensionless power** $\Gamma = \sum_n P_n$ is estimated to be ~ 50



For comparison, analytic studies imply $P_n \simeq \frac{\Gamma}{\zeta(q)n^q}$ with $q = \frac{4}{3}, \frac{5}{3}, 2$ for **cusps**, **kinks**, and **kink-kink collisions**



Stochastic GW Background Induced by Cosmic Strings



The **energy** of **cosmic strings** is converted into the **energy** of **GWs**, and an **SGWB** is formed due to **incoherent superposition**



The **SGWB energy density** ρ_{GW} per unit frequency at the present is

$$\frac{d\rho_{\text{GW}}}{df} = G\mu^2 \int_{t_{\text{ini}}}^{t_0} a^5(t) \sum_n \frac{2nP_n}{f^2} n_{\text{CS}} \left(\frac{2na(t)}{f}, t \right) dt$$



$n_{\text{CS}}(l, t)$ is the **number density per unit length** of **CS loops** with length l at cosmic time t



$a(t)$ is the **scale factor** satisfying $\frac{da(t)}{dt} = a(t)H(t)$ and $a(t_0) = 1$



$H(t)$ is the **Hubble rate** and t_{ini} is the cosmic time when the GW emissions start



The **SGWB spectrum** is commonly represented by

$$\Omega_{\text{GW}}(f) = \frac{f}{\rho_c} \frac{d\rho_{\text{GW}}}{df}, \quad \rho_c \equiv \frac{3H_0^2}{8\pi G}$$

Velocity-dependent One-scale Model



The evolution of the **CS network** can be described using the **velocity-dependent one-scale (VOS) model** [Martins & Shellard, hep-ph/9507335, PRD]



The parameters are the **correlation length** L and the **root-mean-square velocity** v of string segments; the **energy density** of **long strings** is expressed as $\rho = \mu/L^2$



Introducing a **dimensionless quantity** $\xi \equiv L/t$, the evolution equations are

$$t\dot{\xi} = H(1 + v^2)t\xi - \xi + \frac{1}{2}\tilde{c}v, \quad t\dot{v} = (1 - v^2) \left[\frac{k(v)}{\xi} - 2Htv \right]$$

$$\tilde{c} \simeq 0.23, \quad k(v) = \frac{2\sqrt{2}}{\pi}(1 - v^2)(1 + 2\sqrt{2}v^3) \frac{1 - 8v^6}{1 + 8v^6}$$



The solutions converge to **constant values** [Marfatia & YL Zhou, 2312.10455, JHEP]:

$$\xi_r = 0.271, \quad v_r = 0.662, \quad \text{radiation-dominated (RD) era}$$

$$\xi_m = 0.625, \quad v_m = 0.582, \quad \text{matter-dominated (MD) era}$$



This implies that the CS network quickly evolves into a **linear scaling regime** characterized by $L \propto t$

Loop Production Functions

🍊 The **CS loop number density** is given by $n_{\text{CS}}(l, t) = \frac{1}{a^3(t)} \int_{t_{\text{ini}}}^t \mathcal{P}(l', t') a^3(t') dt'$

🍏 Motivated by **numerical simulations** [Blanco-Pillado, Olum & Shlaer, 1309.6637, PRD], the **loop production functions** can be approximated as

$$\mathcal{P}_r(l, t) = \frac{\mathcal{F}_r \tilde{c} v \delta(\alpha_r \xi - l/t)}{\gamma_v \alpha_r \xi^4 t^5}, \quad \text{RD era}$$

$$\mathcal{P}_m(l, t) = \frac{\mathcal{F}_m \tilde{c} v \Theta(\alpha_m \xi - l/t)}{\gamma_v (l/t)^{1.69} \xi^3 t^5}, \quad \text{MD era}$$

🍏 $\gamma_v = (1 - v^2)^{-1/2}$ is the Lorentz factor

🍒 At the **loop production time** t_* , we have

$$l_* = l + \Gamma G \mu (t - t_*), \quad \alpha_r \xi_* \simeq 0.1 \text{ and } \alpha_m \xi_* \simeq 0.18$$

🍏 Adopting $\mathcal{F}_r = 0.1$ and $\mathcal{F}_m = 0.36$, the obtained **loop number densities** in the **RD** and **MD eras** **agrees** with the **simulation results** in the **scaling regime**

Loop Production Functions

🍊 The **CS loop number density** is given by $n_{\text{CS}}(l, t) = \frac{1}{a^3(t)} \int_{t_{\text{ini}}}^t \mathcal{P}(l', t') a^3(t') dt'$

🍏 Motivated by **numerical simulations** [Blanco-Pillado, Olum & Shlaer, 1309.6637, PRD], the **loop production functions** can be approximated as

$$\mathcal{P}_r(l, t) = \frac{\mathcal{F}_r \tilde{c} v \delta(\alpha_r \xi - l/t)}{\gamma_v \alpha_r \xi^4 t^5}, \quad \text{RD era}$$

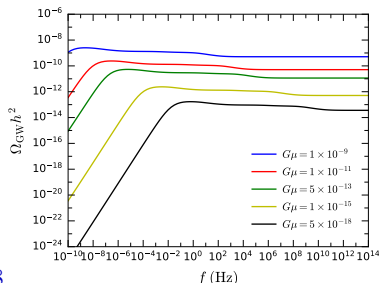
$$\mathcal{P}_m(l, t) = \frac{\mathcal{F}_m \tilde{c} v \Theta(\alpha_m \xi - l/t)}{\gamma_v (l/t)^{1.69} \xi^3 t^5}, \quad \text{MD era}$$

🍏 $\gamma_v = (1 - v^2)^{-1/2}$ is the Lorentz factor

🍒 At the **loop production time** t_* , we have
 $l_* = l + \Gamma G \mu (t - t_*)$, $\alpha_r \xi_* \simeq 0.1$ and $\alpha_m \xi_* \simeq 0.18$

🍏 Adopting $\mathcal{F}_r = 0.1$ and $\mathcal{F}_m = 0.36$, the obtained **loop number densities** in the **RD** and **MD eras** agrees with the **simulation results** in the **scaling regime**

🍏 The **SGWB spectra** in the Λ CDM cosmological model is further calculated



Early Cosmic History



Cosmological observations can **hardly** date back to eras **prior to big bang nucleosynthesis** (BBN)



Various hypotheses beyond the standard cosmic history **predating BBN** are possible, such as an **early matter-dominated (EMD) era**, a **kination-dominated era**, and an **intermediate inflationary era**



Traditional **electromagnetic detection** methods are **ineffective** when the Universe was **opaque to photons**

Early Cosmic History



Cosmological observations can **hardly** date back to eras **prior to big bang nucleosynthesis** (BBN)



Various hypotheses beyond the standard cosmic history **predating BBN** are possible, such as an **early matter-dominated (EMD) era**, a **kination-dominated era**, and an **intermediate inflationary era**



Traditional **electromagnetic detection** methods are **ineffective** when the Universe was **opaque** to **photons**



Nonetheless, **GWs** can **propagate freely** through space, preserving information from the early Universe and reaching us in the present day



We study how the **SGWB spectrum** originated from a preexisting **CS network** is modified by an **EMD era** [SQ Ling & ZHY, 2502.16576, CPC]



Origin of the Early Matter-dominated Era



Consider **dark matter (DM) dilution mechanism** as the origin of the **EMD era**



Thermal production of a **light DM candidate X** with low annihilation cross sections typically results in an **overproduction problem**



DM overproduction can be **mitigated** by **entropy injection** from the **decays** of a **dilutor particle Y** , which **dominates** the Universe for a period, inducing an **EMD era**



Taking the **minimal left-right symmetric model** as an example, where the lightest and next-to-lightest right-handed neutrinos N_1 and N_2 can serve as X and Y



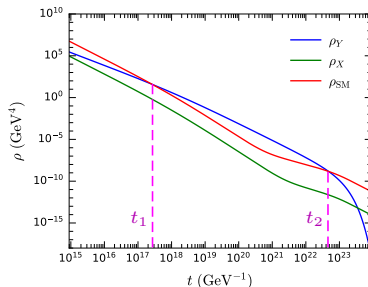
The related Boltzmann equations are

$$\frac{d\rho_Y}{dt} + 3H\rho_Y = -\Gamma_Y\rho_Y$$

$$\frac{d\rho_X}{dt} + 4H\rho_X = yB_X\Gamma_Y\rho_Y$$

$$\frac{d\rho_{SM}}{dt} + 4H\rho_{SM} = (1 - yB_X)\Gamma_Y\rho_Y$$

[Nemevšek & Y Zhang, 2206.11293, PRL]

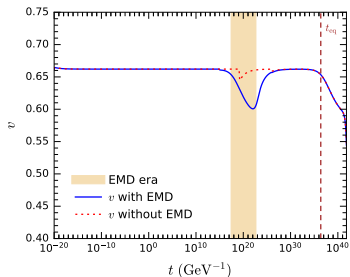
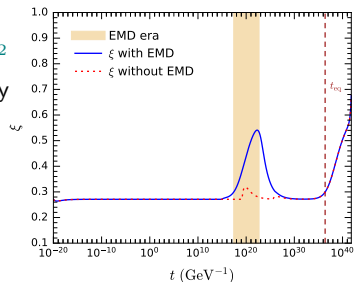
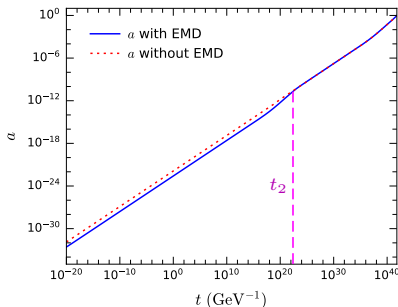


Impact on the Scale Factor and the VOS Parameters

🦎 Compared with the Λ CDM model, the presence of the EMD era reduces the scale factor a before t_2

🐍 $a \propto t^{2/3}$ during an MD era increases more rapidly than $a \propto t^{1/2}$ during an RD era, and a is smaller at the onset of the EMD era to ensure $a(t_0) = 1$

🦎 Moreover, the EMD era introduces a nonscaling effect to the evolution of the CS network



Imprints in the SGWB spectrum



Affected by the **EMD era**, the SGWB spectrum displays a **suppression** at **high frequencies**



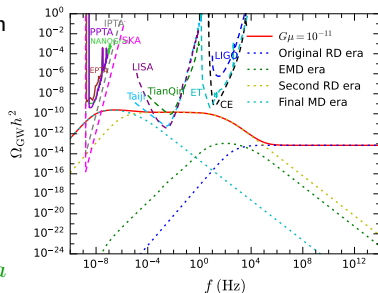
This corresponds to the contributions from CS loops formed in the **original RD** and **EMD eras**



The **lengths** of the generated **CS loops** are **positively correlated** with the **scale factor** a



Since the **EMD era reduces the scale factor** a **before** t_2 , the CS loops with a given **initial length** l , which is related to the **GW emission frequency** by $f_e = 2n/l$, are **formed at a later time**, when the **energy densities** of both CS loops and the emitted GWs are **reduced**



Imprints in the SGWB spectrum



Affected by the **EMD era**, the SGWB spectrum displays a **suppression** at **high frequencies**



This corresponds to the contributions from CS loops formed in the **original RD** and **EMD eras**



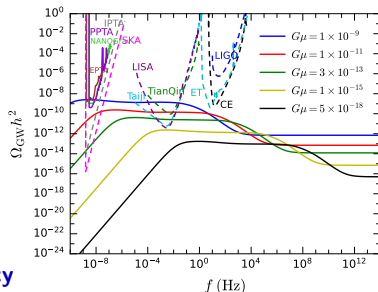
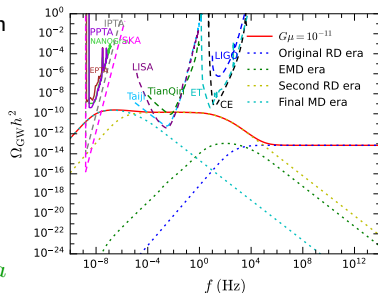
The **lengths** of the generated **CS loops** are **positively correlated** with the **scale factor** a



Since the **EMD era reduces the scale factor** a **before** t_2 , the CS loops with a given **initial length** l , which is related to the **GW emission frequency** by $f_e = 2n/l$, are **formed at a later time**, when the **energy densities** of both CS loops and the emitted GWs are **reduced**



For a **smaller CS tension**, the **loop lifetime** is **extended** and the **average length** at t_2 is **smaller**, causing **suppression** to begin at a **higher frequency**



Summary

- In the early Universe, the **spontaneous breaking** of **symmetries** could lead to **topological defects**, such as **domain walls** and **cosmic strings**
- **Cosmic strings** or **collapsing domain walls** may result in a **stochastic GW background**, which could be probed in GW experiments
- We consider **quantum** and **thermal corrections** to the **effective potential** and explore their impact on the **dynamics** of **domain walls** and the resulting **GW signatures**
- We investigate how an **early matter-dominated era** in cosmic history influences the **dynamics** of **cosmic strings** and the produced **GW spectrum**

Summary

- In the early Universe, the **spontaneous breaking** of **symmetries** could lead to **topological defects**, such as **domain walls** and **cosmic strings**
- **Cosmic strings** or **collapsing domain walls** may result in a **stochastic GW background**, which could be probed in GW experiments
- We consider **quantum** and **thermal corrections** to the **effective potential** and explore their impact on the **dynamics** of **domain walls** and the resulting **GW signatures**
- We investigate how an **early matter-dominated era** in cosmic history influences the **dynamics** of **cosmic strings** and the produced **GW spectrum**

Thanks for your attention!

Friction on the Domain Walls



The interaction between a **domain wall** and the **f fermions** in the **thermal bath** induces **friction** on the wall as it moves in the plasma



The **friction force** per unit area exerted on the **DW** is

$$F_f = \frac{2}{\pi^2} \frac{1}{1 - v_{\text{DW}}^2} \int_0^{+\infty} \int_{-\infty}^{+\infty} R(p_x) \frac{(p_x - \omega v_{\text{DW}})^2}{\omega - p_x v_{\text{DW}}} \frac{1}{e^{\omega/T} + 1} p_{\perp} dp_x dp_{\perp}$$



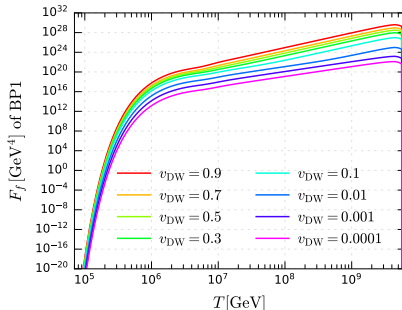
The **reflection probability** $R(p_x)$ can be estimated by considering one-dimensional scattering of a free particle off a step potential



F_f **decreases exponentially** due to the **Boltzmann suppression** at **low temperatures**



The friction is **negligible** when evaluating the **annihilation temperature** T_{ann} for the BPs



Values of T_{ann} , f_{peak} , and $\Omega_{\text{GW}}(f_{\text{peak}})h^2$ for the BPs

	T_{ann} [GeV]	f_{peak} [Hz]	$\Omega_{\text{GW}}(f_{\text{peak}})h^2$
BP1	7.67×10^4	1.28×10^{-2}	2.49×10^{-9}
BP1 w/o f	2.45×10^5	4.08×10^{-2}	2.39×10^{-11}
BP2	7.62×10^{-2}	8.66×10^{-9}	3.49×10^{-12}
BP2 w/o f	1.94×10^{-2}	2.20×10^{-9}	8.31×10^{-10}
BP3	1.19×10^7	1.97	6.82×10^{-8}
BP3 w/o f	1.04×10^8	17.4	1.14×10^{-11}

DM Dilution Mechanism

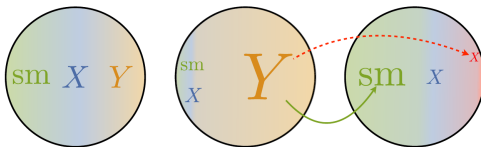
🎯 A **long-lived dilutor** Y has mass m_Y **much larger** than m_X can effectively address the **overproduction problem** of X particles

🌂 First, during the **RD era**, both Y and X particles **decouple relativistically** at a similar temperature, resulting in comparable yields, $Y_Y \simeq Y_X$

⑧ Second, because of $m_Y \gg m_X$, Y particles become **nonrelativistic** at a relatively high temperature, while X particles remain **relativistic**

🎃 Consequently, Y particles quickly **dominate** the energy density of the Universe, initiating an **EMD era**

🎈 Finally, when the **lifetime** of Y particles comes to an end, they **decay** into SM particles and X particles, **injecting entropy** and consequently **diluting** the **energy density** ρ_X of X particles



[Nemevšek & Y Zhang,
2206.11293, PRL]

Minimal Left-right Symmetric Model



The **DM candidate** is $X = N_1$, and the **dilutor** is $Y = N_2$, which undergoes a three-body decay mediated by a **right-handed gauge boson** $W_R^\pm$ into two charged leptons $\ell\ell'$ and one N_1



The related **right-handed charged current interactions** are described by

$$\mathcal{L}_1 = \frac{g}{\sqrt{2}} W_R^\mu \left(\sum_{i=1}^2 \bar{N}_i \gamma_\mu V_{\text{PMNS}}^{R\dagger} \ell_R + \bar{u}_R \gamma_\mu V_{\text{CKM}}^R d_R \right) + \text{H.c.}$$



N_2 **decay channels** include $N_2 \rightarrow N_1 \ell\ell'$, $N_2 \rightarrow \ell q \bar{q}'$, and $N_2 \rightarrow \ell W$



Benchmark parameters used in the previous slides:

$$m_{N_2} = 200 \text{ GeV}, \quad m_{N_1} = 6.5 \text{ keV}, \quad m_{W_R} = 5 \times 10^7 \text{ GeV}, \quad \tan \beta = 0.5$$

$$\Gamma_{N_2} = 2.22 \times 10^{-23} \text{ GeV}, \quad B_X = 4.41 \times 10^{-3}, \quad y = 0.35,$$



B_X is the **branching ratio** of the decay channel $N_2 \rightarrow N_1 \ell\ell'$



y is the **energy fraction** carried away by the X particle from the Y particle

Effects of the Dilutor Decay Width and Mass



A **smaller dilutor decay width Γ_Y** corresponds to a **longer duration** of the **EMD era**, leading to **stronger suppression** effects at high frequencies



A **larger dilutor mass m_Y** implies that the **EMD era** occurs **earlier**, and hence a **higher frequency** at which the suppression of the GW spectrum commences

